

Question number	Answer	Additional guidance	Mark
8(a)(i)	GUG	ACCEPT guanine uracil guanine / CAC / cytosine adenine cytosine IGNORE val / valine	(1)

Question number	Answer	Additional guidance	Mark
8(a)(ii)	substitution	DO NOT ACCEPT frameshift / deletion / addition / insertion IGNORE {gene / point} mutation	(1)

Question number	Answer	Additional guidance	Mark
8(a)(iii)	An answer that includes the following points: - the R groups (of these two amino acids) have different {properties / bonding} (1) - glu may have repelled polar groups on other haemoglobin molecules (1) {val / hydrophobic R group / hydrophobic part} might form other (hydrophobic) interactions (with other haemoglobin molecules) (1) (part of haemoglobin containing) val (R group) turns away from {water / cytoplasm} (1)	ACCEPT amino acids have different properties ACCEPT bonds / Van der Waals DO NOT ACCEPT incorrect named bonds ACCEPT repels water DO NOT ACCEPT blood / plasma	(3)

Question number	Answer	Additional guidance	Mark
8(b)	- number of non-affected babies calculated (1) $0.002 : 1 / 2 \times 10^{-3} : 1$ (1)	(140 million - 305 800 =) 139 694 200 0.002189 ACCEPT 0.0022 / 0.00219 / 0.002189 ACCEPT 1 : 457 / 456.8 NB 1 mark for 1 : 458 / 1 : 457.8 0.002184 : 1 (0.002 / 0.0022 / 0.0218) Bald answer = 2 marks	(2)

Question number	Answer	Additional guidance	Mark
8(c)(i)	- values read correctly from the graph (1) 3.3 / 3.4 (1)	ACCEPT 4.4 / 4.9 and 5.7 / 5.8 / 7.7 / 7.8 ACCEPT ecf for correct subtraction using {4.4 / 7.7 / 7.8} and one incorrect value Bald answer = 2 marks	(2)

Question number	Answer	Additional guidance	Mark
8(c)(ii)	An answer that includes the following points: • person without the disease has higher saturation than person with the disease (at all partial pressures of oxygen) (1) • there is {greater variability / wider range} (in the saturation) of a person with the disease than a person without the disease (at a particular partial pressure of oxygen) (1)	NB accept converse throughout but all conclusions must be comparative ACCEPT refs to affinity (for oxygen) IGNORE graph shifted	(2)

Question number	Answer	Additional guidance	Mark
8(c)(iii)	• 120 (days) (1)		(1)

Question number	Answer	Additional guidance	Mark
8(c)(iv)	An explanation that includes three of the following points: • (change in structure of haemoglobin) haemoglobin {binds / carries} less oxygen (1) • (shape of red blood cell) smaller surface area so less oxygen <u>diffuses</u> in / red blood cells get lodged in blood vessels preventing flow of blood to cells (1) • therefore less oxygen to {cells / tissues} so {less available for (aerobic) <u>respiration</u> / switch to <u>anaerobic respiration</u>} (1) • credit an example of why less oxygen to cells could be fatal (1)	NB accept converse where appropriate ACCEPT named cell / tissue e.g. heart attack, stroke, sepsis, infection	(3)